HEALTHCARE AI – HYBRID GRAPH RAG CHATBOT

Project Features & Technology Scope

PROJECT OVERVIEW

A healthcare-focused AI chatbot where users can select specialized agents such as Cardiology, Gynaecology, Neurology, etc. Users can upload relevant documents and ask questions based on their uploaded knowledge.

CORE FEATURES

1. Specialized Healthcare Agents

• Domain-specific AI agents such as Cardiology and Gynaecology

• Each agent has its own healthcare context and instructions

• Easy to add new healthcare domains in the future

2. Multi-Format Document Upload

• PDF, Word, Excel, PowerPoint and TXT files

• Documents are processed and added to the agent's knowledge base

3. Hybrid Graph RAG

The chatbot will use three complementary retrieval approaches:

• Semantic/vector search – understands the meaning of questions

• Keyword/BM25 search – finds exact medical terms and phrases

• Knowledge Graph (Neo4j) – understands relationships between diseases, symptoms, medicines, treatments and other entities

The retrieved information will be combined and ranked before being provided to the AI model, helping produce more relevant and grounded responses.

4. Intelligent Chatbot

• Answers questions using uploaded documents and domain knowledge

• Maintains conversation context

• Provides document/source references where applicable

• Designed to reduce unsupported or hallucinated answers

5. AI-Powered Tools

• Mermaid diagrams for medical processes and relationships

• KaTeX for formulas and scientific notation

• Charts and visual representations from structured data

• Structured tables and summaries

OPTIONAL ADD-ONS

• Understanding charts and images inside PDFs

• Voice input for asking questions

• Ask questions across multiple uploaded documents

• AI-generated follow-up question suggestions

TECHNOLOGY STACK

Frontend

- React + TypeScript
- Tailwind CSS
- Mermaid
- KaTeX
- Charting library

Backend

- FastAPI + Python

AI & RAG

- Configurable LLM such as Gemini / Groq
- Embedding-based semantic search
- BM25 keyword search
- Neo4j Knowledge Graph
- Hybrid retrieval + result fusion + reranking

Database

- PostgreSQL – users, chats, documents and application data
- Qdrant – vector/sparse retrieval
- Neo4j – healthcare entities and relationships

The architecture will be modular so that new healthcare agents, document types, AI models and tools can be added without rebuilding the complete platform.